

Bank Loan Default Risk Analysis

1. Project Overview

The Bank Loan Default Risk Analysis project aims to identify key factors that influence borrower default behavior and evaluate the overall risk profile of a loan portfolio. Using SQL for data analysis and Tableau for visualization, the project examines borrower demographics, financial characteristics, and loan attributes to uncover patterns associated with loan defaults.

The insights generated from this analysis can support financial institutions in improving underwriting decisions, reducing credit risk, and enhancing portfolio performance.

2. Dataset Overview

The dataset contains information related to loan applicants, their financial background, employment details, loan characteristics, and repayment outcomes.

Key Variables Included:

Borrower Information

- Age
- Income
- Education
- Employment Type
- Marital Status
- Dependents

Credit & Financial Information

- Credit Score
- Number of Credit Lines
- Debt-to-Income (DTI) Ratio
- Mortgage Status
- Co-Signer Status

Loan Information

- Loan Amount
- Interest Rate
- Loan Term
- EMI

- Loan Purpose

Target Variable

- Loan Default Status (Default / Non-Default)

Derived Features Created

To improve analysis and reporting, additional fields were generated:

- Credit Score Band
- DTI Band
- EMI to Income Ratio
- Tenure Stability

These derived metrics helped segment borrowers into meaningful risk categories.

3. Project Objectives

1. Measure the overall loan default rate.

	total_loans	default_loans	default_rate
1	255347	29653	11.61

2. Analyze default behavior across loan purposes.

	loan_purpose	total_loans	default_loans	default_rate
1	Business	51298	6323	12.33
2	Auto	50844	6041	11.88
3	Education	51005	6038	11.84
4	Other	50914	6002	11.79
5	Home	51286	5249	10.23

3. Evaluate the impact of credit scores on repayment performance.

	credit_score_band	total_loans	default_loans	default_rate
1	Poor	130223	16245	12.47
2	Fair	41847	4784	11.43
3	Good	32073	3409	10.63
4	Excellent	51204	5215	10.18

4. Assess the relationship between DTI ratio and default risk.

	dti_band	total_loans	default_loans	default_rate
1	Very High	94334	11526	12.22
2	High	63644	7525	11.82
3	Moderate	63856	7129	11.16
4	Low	33513	3473	10.36

5. Examine employment stability and borrower reliability.

	emi_burden	total_loans	default_loans	default_rate
1	Low	255347	29653	11.61

6. Compare risk levels among different borrower segments.

	tenure_stability	total_loans	default_loans	default_rate
1	Unstable	50935	8270	16.24
2	Stable	204412	21383	10.46

7. Identify high-risk customer profiles.

	has_co_signer	total_loans	default_loans	default_rate
1	0	127646	16423	12.87
2	1	127701	13230	10.36

	has_mortgage	total_loans	default_loans	default_rate
1	0	127670	15761	12.35
2	1	127677	13892	10.88

8. Provide recommendations to reduce future defaults.

	credit_score_band	dti_band	tenure_stability	total_loans	default_loans	default_rate
1	Poor	Very High	Unstable	9501	1736	18.27
2	Poor	Moderate	Unstable	6412	1116	17.40
3	Poor	High	Unstable	6514	1118	17.16
4	Fair	High	Unstable	2017	341	16.91
5	Good	Very High	Unstable	2417	408	16.88
6	Fair	Very High	Unstable	3155	500	15.85
7	Good	High	Unstable	1617	252	15.58
8	Fair	Moderate	Unstable	2138	328	15.34
9	Fair	Low	Unstable	1082	161	14.88
10	Poor	Low	Unstable	3420	507	14.82

4. Methodology

Data Preparation

- Reviewed dataset structure and attributes.
- Standardized column naming conventions.
- Created additional analytical fields for segmentation.
- Prepared the dataset for SQL querying and Tableau visualization.

SQL Analysis

SQL queries were used to:

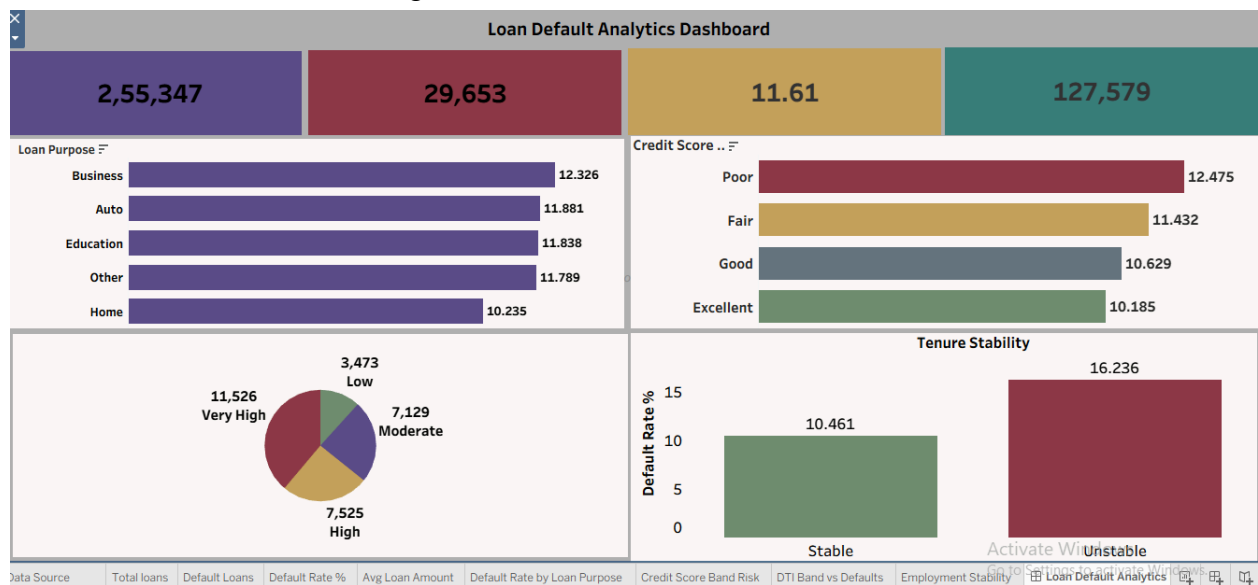
- Calculate default rates.
- Analyze borrower demographics.

- Evaluate loan-purpose risk.
- Measure credit-score performance.
- Assess DTI and employment-related risk factors.

Tableau Dashboard Development

A single-page interactive dashboard was developed containing:

- Total Loans
- Default Loans
- Default Rate (%)
- Average Loan Amount
- Default Rate by Loan Purpose
- Credit Score Risk Analysis
- DTI Risk Analysis
- Employment Stability Analysis
- Interactive Filters and Segmentation



5. Key Findings

The analysis revealed several important trends:

- Borrowers with weaker credit profiles generally exhibited higher default rates.
- Higher debt burdens were associated with increased repayment risk.
- Certain loan purposes demonstrated greater default tendencies.
- Employment stability appeared to influence repayment performance.
- Combining multiple risk factors resulted in significantly higher default probabilities.

6. Recommendations

Based on the analysis, the following actions are recommended:

1. Strengthen credit score screening during loan approval.
 2. Apply stricter monitoring for borrowers with high DTI ratios.
 3. Incorporate employment stability into underwriting decisions.
 4. Encourage the use of co-signers for higher-risk applicants.
 5. Implement risk-based pricing strategies.
 6. Develop early-warning systems for high-risk borrower segments.
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7. Conclusion

This project demonstrates how data analytics can be leveraged to understand loan default behavior and improve lending decisions. By combining SQL-based analysis with Tableau visualization, meaningful insights were generated regarding borrower risk, credit quality, and portfolio performance. The findings can assist financial institutions in reducing default rates while maintaining sustainable loan growth.